Requirements Specification Document

SNOMED_CT Implementation

Versie 0.8

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Document Information

Document history

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Document validation

1. General

1.1 Introduction

The European Health Data Space (EHDS) Regulation aims to establish a common framework for the use and exchange of electronic health data across the European Union. This regulation strengthens citizens' access to their personal electronic health data and increases their control over how that data is used.
To enable the smooth implementation of the EHDS Regulation, there is a need for a single European technical FHIR standard for each message type, including the exchange of essential patient information (e.g., the SumEHR). In addition, each relevant data element within a message type requires a single European structured semantic value set. These value sets should preferably align with internationally recognized standards such as SNOMED CT, with mappings to classifications and terminologies such as ICPC-3, ICD-11, ICF, and ICHI.

The **SNOMED CT Implementation** project aims to define a framework that meets both the requirements of the EHDS Regulation and the needs of clinical practice. The project also aligns with the objectives of the national action plan for the digitalization and standardization of health data. Within this context, the **Belgian Integrated Health Record (BIHR)** plays an important role. The BIHR aims to support the harmonization of health data through standardized clinical data sets, data models, terminologies and artifacts. The implementation of SNOMED CT is an essential building block in this effort, enabling semantic interoperability and supporting the consistent recording, exchange, and reuse of health information.
Within the project, two complementary workstreams are distinguished: a technical workstream and a clinical workstream.
The clinical workstream focuses on the needs and expectations of healthcare professionals. Through consultations and surveys involving clinicians, the key business needs from the healthcare field have been identified. Some of these needs overlap with technical requirements, such as the availability of appropriate tooling and the capacity to seamlessly integrate complementary tooling functionalities into the eHR systems.
The technical workstream focuses on implementing the necessary technical infrastructure and standards that support semantic interoperability. This includes, among other things, the installation and configuration of a local terminology server, the implementation of SNOMED CT, and the support of European FHIR profiles and value sets required for the compliant exchange of health data within the EHDS context.
Both workstreams are closely interconnected and must be carried out in a coordinated manner to ensure the successful implementation of SNOMED CT, support the BIHR initiative, and contribute to achieving the objectives of the EHDS Regulation.

1.2 Purpose of the document

This Requirements Specification Document aims to establish a uniform set of functional, technical, and organizational requirements for the implementation SNOMED CT within healthcare information systems. The document serves as a reference framework for healthcare organizations, software vendors, and other stakeholders seeking to implement SNOMED CT in alignment with national objectives, the Belgian Integrated Health Record (BIHR), and the requirements of the European Health Data Space (EHDS).

The requirements described in this document are the basis for assessing the maturity and compliance of a SNOMED CT implementation. They serve as objective criteria for the registration of the “snomedization” of Electronic Health Records and related healthcare applications.
Compliance with these requirements is necessary to demonstrate that the required technical and clinical conditions are met in order to:

1/ be registered; and
2/ be prepared for the cross-border exchange of Patient Summary data as mandated by the European Health Data Space (EHDS).

1.3 Objectives & Categories

The implementation of SNOMED CT within Electronic Health Records (EHRs) is based on **four strategic objectives** that together form the foundation of the technical implementation of the project.

- Terminology Services & Provisioning
- Data entry
- Storage
- Interoperability

These objectives share a common goal: to promote the correct, consistent, and sustainable use of SNOMED CT in clinical practice. These objectives are an absolute prerequisite for obtaining high-quality data; semantic interoperability between healthcare systems is also supported, which is essential for national and European data exchange.

To achieve these objectives, a set of functional, technical, and organizational requirements has been defined. These requirements describe the minimum expectations that an implementation must meet to ensure a compliant and high-quality use of SNOMED CT within EHR systems.
The requirements are grouped into different categories, each representing a specific and essential characteristic of a successful SNOMED CT implementation. These categories are based on the implementation principles and best practices established by SNOMED International.

1.3.1. Categories

Versioning It is important that suppliers work with valid editions, versions, and derivative products (documents, subsets, map-pings, etc. that reference SNOMED CT concepts) of SNOMED CT that are relevant within the national context. This is also important when taking into account the native language(s) of national healthcare systems, where relevant and where the native language version is available.
Maintenance The ability to incorporate changes from the international and national editions of SNOMED CT is essential for a successful long-term implementation. The Belgian SNOMED CT Edition is released monthly, and National Release Centers also publish their national editions on a monthly basis. These updates contain important changes that may result from advances in medical knowledge or from corrections made following feedback from the user and clinical communities.
User interface The success of an implementation of SNOMED CT is often driven by the quality and approach that has been employed in the development of the user interface. This is where end users will interact and use SNOMED CT and a poor implementation will have a considerable impact on the quality of data entered.
Data Capture

The search and selection functionality within the user interface plays a crucial role in the daily use of the system. A well-designed and efficiently used search function enables users to find the right information, concepts, or data more quickly, significantly reducing the time required to perform tasks.
Storage The way in which SNOMED CT is stored in the Electronic Patient Record (EPR) and effectively reused in clinical practice for the benefit of both the clinician and the patient is another factor that can determine whether an implementation is successful. The use of SNOMED CT concept identifiers is a mandatory requirement.
Analytics & reporting One of the key drivers for the implementation of SNOMED CT is the structuring of data and the use of its ontology to derive clinical insights from patient data that would otherwise not be available in unstructured data.
Interoperability An important aspect of SNOMED CT implementations is how interoperability is implemented. SNOMED CT is a very powerful tool to ensure the safe, unambiguous transfer of patient data between organizations and the use of SNOMED CT concept identifiers is at the foundation of this.

2. Requirements

2.1 Objective 1 - Terminology Services & Provisioning

The eHR must include terminology services that are FHIR and ECL enabled, and it must support the correct and most up-to-date version of the semantic standard, including all subsets, mappings, and available translations.

2.1.1 Terminology Services

N°	**TSP1**
Summary	FHIR enabled services
Description	FHIR terminology API: systems must natively support the HL7 FHIR terminology service standard. They must execute : $expand (for value sets), $validate-code (for validation), and $lookup (for concept details) operations.
Category	/

N°	**TSP2**
Summary	ECL enabled services
Description	The system must not treat clinical codes as flat lists, it uses the logical hierarchy and data structure of SCT. This is best achieved by supporting SCT ECL query, that is accessible via a user friendly interface
Category	/

2.1.2 Terminology Provisioning

N°	**TSP3**
Summary	Incorporate the last/most recent Belgian Edition of SNOMED CT, including all published subsets and maps, via automated syndication ingestion
Description	The Belgian Edition of SCT is updated monthly in the repository of the eHR system via an automated, scheduled background synchronization to check the central NTS for updates. This process must run asynchronously without breaking or interrupting any clinical workflows. Ensure that the version of the given SNOMED CT edition being used within the solution is the last/most recent version release by the BE NRC. Components that may change between versions may include: ○ **Concepts**: new concepts added, concepts made inactive. ○ **Descriptions**: new descriptions added, descriptions made inactive, changes of case sensitivity or acceptability, … ○ **Relationships**: new stated or inferred relationships ○ **Refsets**: new refsets, new members to a refset, members made inactive within a refset, deprecated refsets Latest/most recent version: release version -1 (monthly releases).
Category	/

N°	**TSP4**
Summary	Multilanguage accommodation
Description	Clinical data must be presented in the appropriate language dialect dynamically using the Language Reference Sets to query the locally validated terms. Belgian edition includes a Dutch, French and German translation.
Category	/

N°	**TSP5**
Summary	Provide a reporting/feedback mechanism
Description	The system will provide user friendly functionalities to assist users in reporting missing or erroneous content regarding concepts, translations, etc. The missing content can be easily reported/flagged by the end user so it can be picked up by terminologists to report this to the Belgian NRC Likewise errors and corrections must be easy to report directly within the eHR system. The feedback mechanism does not need to be directly integrated with the NRC portal. However, it is important that users can submit feedback on SNOMED CT content, translations, mappings, or other terminology-related aspects in a clear and accessible manner. The NRC portal serves as the official and sole channel for the registration and management of such feedback. Link to the portal : https://apps.health.belgium.be/terminology-portal/snomed_ct_requests/nl

	https://apps.health.belgium.be/terminology-portal/snomed_ct_requests/fr
Category	Versioning

2.2 Objective 2 – Data entry

Data entry is made as easy as possible: the user-friendly search and selection of terms/concepts is supported by the necessary eHR functionalities and complementary tools, with no learning curve required for users (e.g.tools can assist in converting free text to SNOMED CT code suggestions to be validate by the end user).

2.2.1 Data capture

A powerful search functionality within the Electronic Health Record (EHR) is an essential prerequisite for the successful implementation of SNOMED CT. Healthcare professionals must be able to quickly and easily find the appropriate concepts in order to record clinical information accurately and consistently.
A well-designed search function not only improves usability but also contributes to a more efficient workflow, reduces documentation burden, and supports higher-quality data capture.
It is therefore important that the search functionality supports users through relevant search results, intelligent search algorithms, synonyms, abbreviations, and other mechanisms that facilitate the selection of the appropriate SNOMED CT concepts. An efficient search experience promotes user adoption of SNOMED CT and directly contributes to consistent and high-quality clinical data recording.

To illustrate possible search functionalities and user experiences, reference can be made to the following demonstration: https://ihtsdo.github.io/snomed-ui-examples/

N°	**U001**
Summary	Overview of set of basic search functionalities
Description	This overview is a set of basic search functionalities designed to help users find the appropriate SNOMED CT code more quickly and efficiently, directly from within the eHR The focus is on simplifying the search and data entry process, enabling both experienced and less experienced users to consistently identify relevant terms. The following requirements for the “data capture” functionality outline the minimum capabilities needed to ensure a user-friendly, performant, and effective search experience.
Category	User Interface / Search & Select

Summary	Description
As the user types each character into a SNOMED CT coded data element, suggest a list of relevant concepts but limit the selection of concepts	As the user types each character into a SNOMED CT coded data element, limit the selection of concepts to those with a preferred or acceptable term that matches the characters types (using a ‘word prefix any order’ algorithm), and use auto-complete when only one option is available for selection.
The eHR system is capable of understanding all variants of a concept description as a search term	The system must be able to understand all variants of a term, such as the plural form and/or the feminine form

The eHR system is capable of word segmentation (decompounding)	When a clinician types “pro hip” the system should retrieve “hip prosthesis”
Support "fuzzy logic" or "Google style” searches (interpret spelling mistakes).	
Support searching for SNOMED CT in the Belgian language reference sets.	Support searching for SNOMED CT concepts using any term that is preferred or acceptable in the Belgian language reference set.
Searching requires a minimum of 3 characters	The system will require a minimum search string of three characters, and the search will only be triggered once the user has entered a three-character term, excluding whitespaces or blank characters.
Search independent of the order of search tokens	The system will perform searches independent of the order of search tokens (e.g., "Skin Cancer" and "Cancer Skin" will return the same results.
The system will default to partial matching in search results and will not require the entry of wildcard characters	The system will default to partial matching in search results and will not require the entry of wildcard characters.
The system will default to returning search results in ascending order of term length.	When displaying the list of possible matches, display the concept with the shortest matching term first.
The system will facilitate real-time progressive matching of term results.	The system will facilitate real-time progressive matching of term results.
The system will use context specific subsets (per medical specialty) to facilitate concept retrieval by the clinician	The system will search first in the context specific subset before browsing the entire SNOMED Ct ontology to fast(er) retrieve relevant concepts according to the context of the end user

N°	**U002**
Summary	The system will enable searches using Concept ID
Description	The system will enable searches using Concept ID and Description ID (optional) It must be possible to search using both the **Concept ID** and the **Description ID** (optional). This enables users to quickly locate a specific concept. Search results should be presented in the language configured for the logged-in user or healthcare practice.
Category	Search & select

N°	**U003**
Summary	Following the return of search results, the system will permit users to browse through IS A relationships.
Description	Based on the search results, users should be able to further navigate the SNOMED CT hierarchical structure, in particular through the **IS-A relationships** between

	concepts. This enables users to better understand the context of a concept and to explore related parent and child concepts. Such navigation may be supported by graphical views, such as hierarchical trees or concept diagrams, as well as by additional help and guidance functionalities.
Category	Search & select

N°	**U004**
Summary	Allow searching and selection of only those concepts from SNOMED CT that have been specifically bound to that data element
Description	Allow searching and selection of only those concepts from SNOMED CT that have been specifically bound to that data element. (terminology binding) The system shall restrict searches and selections to the SNOMED CT hierarchy or branch that has been explicitly associated with the relevant data element. This ensures that users can only select concepts that are clinically and semantically appropriate for the context in which the data is being recorded. Examples of search and navigation capabilities that support efficient concept selection are available through the SNOMED CT search demonstration: https://ihtsdo.github.io/snomed-ui-examples/
Category	Search & select

N°	**U005**
Summary	Allow data capture via free text when concepts can't be found
Description	The eHR system must provide the possibility to register data for which concepts or descriptions are currently missing. The missing content can be easily reported/flagged by the end user so it can be picked up by terminologists to report this to the Belgian NRC (cf. **TSP5**)
Category	Versioning

N°	**U006**
Summary	Show user preferred display term for each SNOMED CT concept.
Description	Upon selection of a concept, use the term entered by the user to display the selected concept;
Category	User Interface

N°	**U007**
Summary	The eHR system shall display the most frequently selected concepts.
Description	The system shall provide functionalities that support the rapid retrieval of commonly used concepts, such as favorites lists, recently selected concepts, context-specific suggested concepts, personalized preferences**, or** other mechanisms that reflect the healthcare professional's usage patterns. This enables users to identify and select the appropriate SNOMED CT concepts with minimal search effort, thereby improving both efficiency and data quality.

Category	User Interface

N°	**U008**
Summary	The eHR is capable of integrating tools/solutions facilitating the search and capture of clinical data.
Description	The eHR is capable of integrating tools & solutions facilitating search and realtime capture of clinical data to suggest possible SNOMED CT encoding for selection and confirmation by the user. (e.g. AI tools)
Category	User Interface

2.3 Objective 3 - Storage

Data shall be captured at the source using the semantic standard code within the EHR. This may be facilitated by a proprietary interface terminology or coding system, provided that the corresponding standard code is also stored. This is required to create and maintain a semantically meaningful patient record containing understandable data in the agreed language, thereby enabling the data to be queried through an agreed query language (ECL).
To be completed with information model requirements: relations between concepts, concepts with “container” function, grouping of concepts, etc.

2.3.1 Storage

N°	**S001**
Summary	Must store SNOMED CT concept identifiers in the health records
Description	Must store SNOMED CT concept identifiers in the health records Store the SNOMED CT concept identifier (or SNOMED CT expression) together with the term selected by the user in the EHR for SNOMED CT coded data element. Storing only the Description ID or free-text labels is a critical failure point.
Category	Storage

N°	**S002**
Summary	Ensure that the context of each SNOMED CT concept identifier or expression is clearly represented.
Description	The clinical meaning of a recorded concept must be represented explicitly and unambiguously. The data model and terminology usage should ensure that the actual clinical concept is captured rather than relying on qualifiers or status indicators alone.
Category	Storage

N°	**S003**
Summary	SNOMED CT component (concepts, descriptions, relationships) status handling
Description	Systems must prevent the storage of deprecated or inactive concepts. The system’s logic must read SCT historical relationship files to automatically point old codes to their active replacements. Deprecated concepts/codes are not deleted and remain visible to the end user of the eHR system. Ensure that all inactivated concepts, descriptions and relationships remain available (and visible to the end user if wanted) to support querying over historical clinical data.
Category	Storage

2.3.2 Analytics & Reporting

N°	**S004**
Summary	Use maps from SNOMED CT to national and international classifications including legacy and current data.
Description	Use maps from SNOMED CT to national and international classifications to enable clinical reuse, analysis and reporting of data, including legacy and current data. The system shall have the capacity to take in maps to other terminologies / classifications where available such as ICD-10, ICPC2, ICHI (NIHDI nomenclature) LOINC, Orphanet codes, and make the maps available in the relevant contexts. This may include on-screen display and inclusion in generated documents, reports and data extracts
Category	Analytics & Reporting

N°	**S005**
Summary	Use maps from local code systems to SNOMED CT only for legacy
Description	Use maps from local code systems to SNOMED CT to enable ad hoc querying, reporting and analysis of legacy clinical data.
Category	Analytics & Reporting

N°	**S006**
Summary	Use SNOMED CT's defining relationships in the determination of the correct results or outcomes.
Description	Applications supporting data reporting, extraction and/or clinical decision support rules shall (where appropriate) use SNOMED CT's defining relationships in the determination of the correct results or outcomes.
Category	Analytics & Reporting

2.4 Objective 4 - Interoperability

Data entered into the system shall be reusable for data exchange (via the relevant care sets and FHIR profiles) as well as for secondary use, without requiring healthcare professionals to enter the same information more than once.

N°	**I001**
Summary	Use SNOMED CT to share data, based on national data standards and populate SNOMED CT coded data elements to exchange.
Description	Use SNOMED CT to populate (automatically) SNOMED CT coded data elements for all relevant message exchanges.
Category	Interoperability

3. Glossary

Term	Comment	
EPD		
EMD		
Concept		
Refset		
Subset		
Variant		
ECL		

3.1 Useful Links

Search Demo : https://ihtsdo.github.io/snomed-ui-examples/

Link to the portal :
https://apps.health.belgium.be/terminology-portal/snomed_ct_requests/nl
https://apps.health.belgium.be/terminology-portal/snomed_ct_requests/fr

3.2 Github